DO NOT WRITE IN THIS AREA

14 Simplify fully $\dfrac{x^2 + 3x + xy + 3y}{x^2 - y^2}$

$x^2 + 3x + xy + 3y = x(x + 3) + y(x + 3)$

1 mark
Factor the numerator by grouping.

$= (x + y)(x + 3)$

$x^2 - y^2 = (x + y)(x - y)$

1 mark
Difference of squares.

$$\frac{(x + y)(x + 3)}{(x + y)(x - y)} = \frac{x + 3}{x - y}$$

.....

(Total for Question 14 is 3 marks)

1 mark
Cancel the common factor (x + y); original restrictions $x \neq \pm y$ remain.

DO NOT WRITE IN THIS AREA